HDN-VDO
SW2000TSN Training Handbook

Automotive Network with TSN Solutions

10BASE-T1S/100BASE-T1/1000BASE-T1 and 2.5/5/10GBASE-T1

Media Converter → Switch → TSN → AVB

HDN-VDO Intelligent Technologies Co., Ltd

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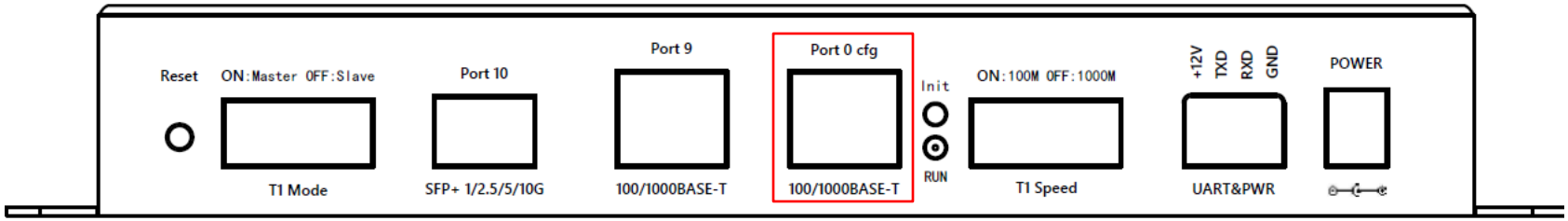
For SW2000 TSN Series Quick Start

SW2000M TSN & SW2000H TSN

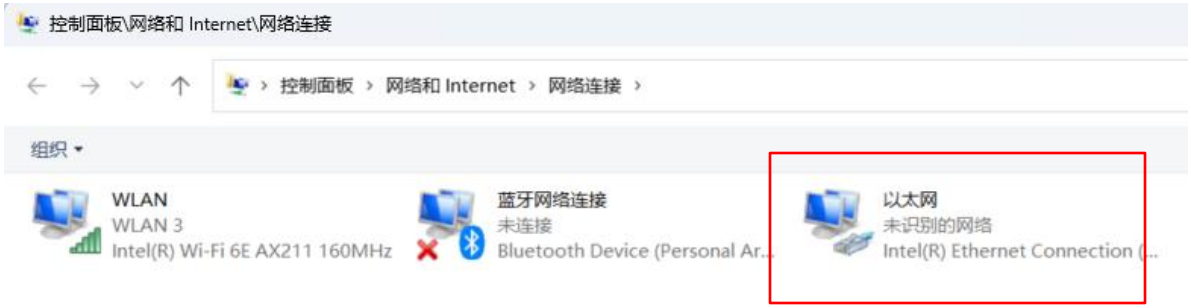
SW2000R TSN

SW2000G-M12X TSN

How to use SW2000TSN Config Tool



Connect Port 0 to your computer, make sure link up 1Gbps.



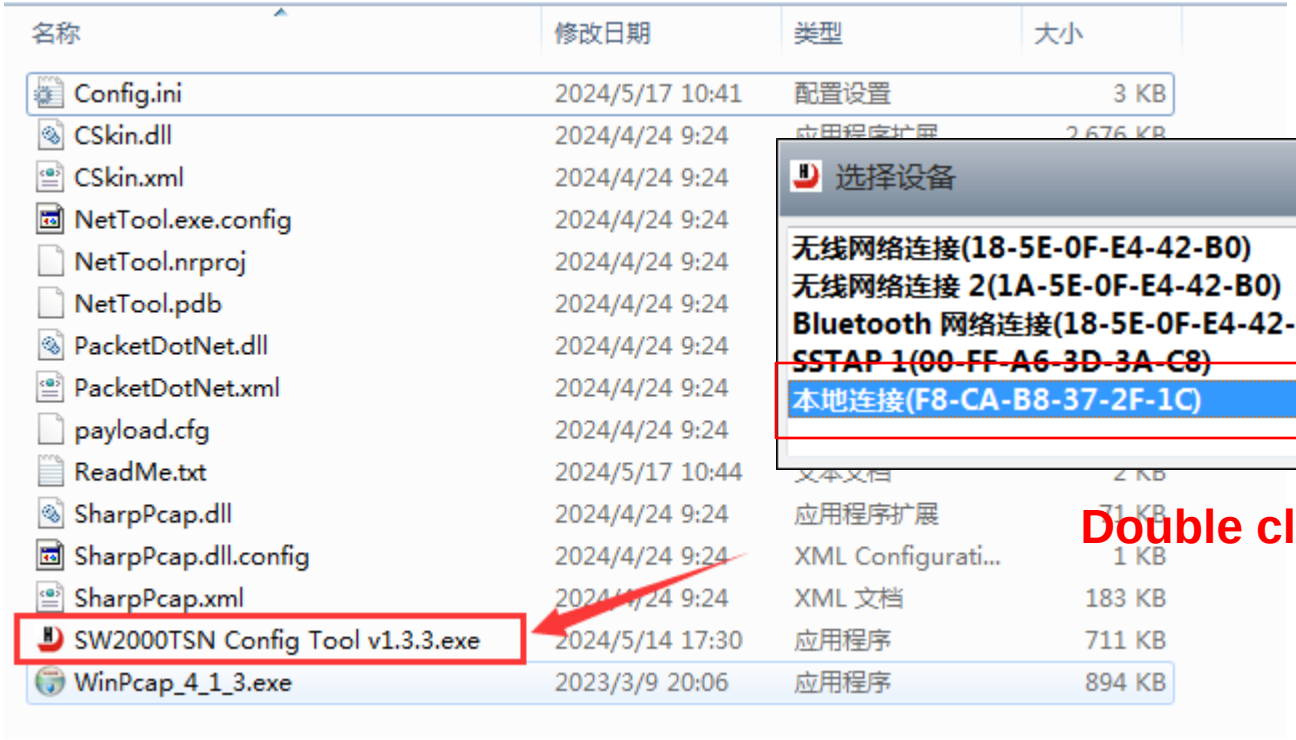
How to use SW2000TSN Config Tool

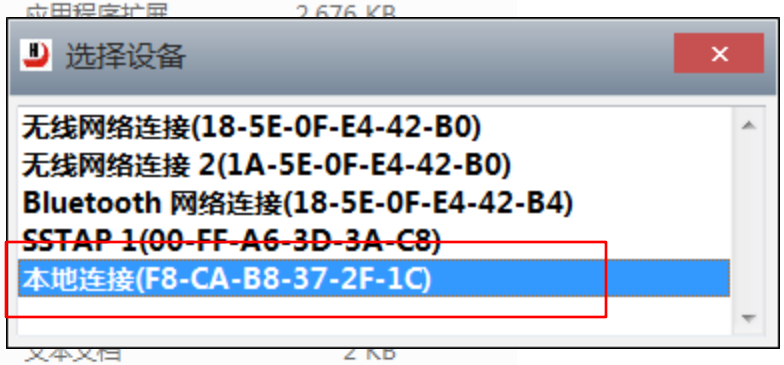
Please install wireshark 3.4.2 first.

Download: https://wwm.lanzouh.com/iK3eQtjvfti

Password: d5tec

Please double click to open the SW2000TSN Config Tool v1.33.exe



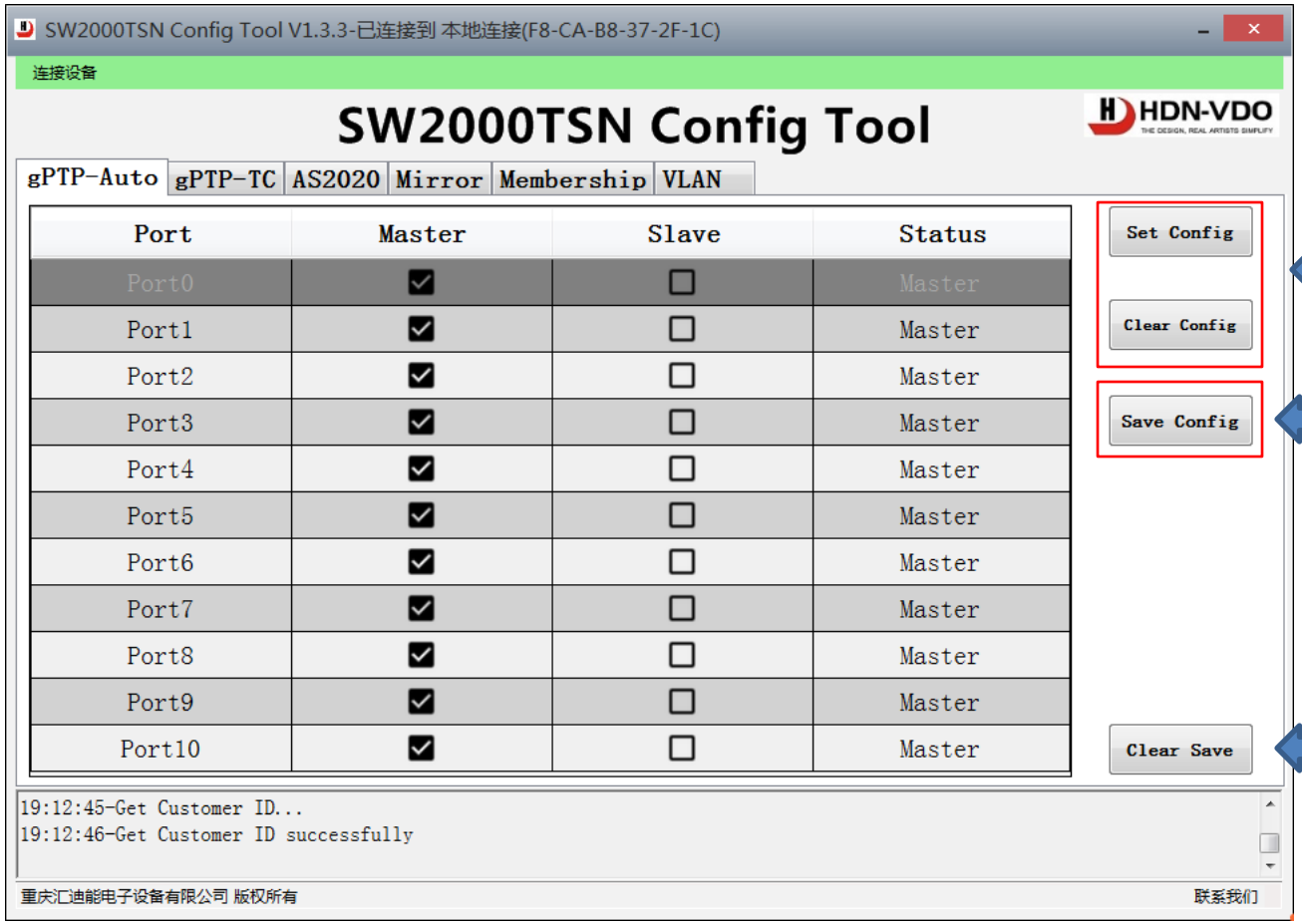


Double click network adapter

*** If you got a error of DLL “wpcap” not found, please double click WinPcap_4_1_3.exe and install it.**

How to use SW2000TSN Config Tool

Select one port as the Slave clock. The remaining ports will be gPTP master clocks and can provide timing to endpoint.



For debug use only

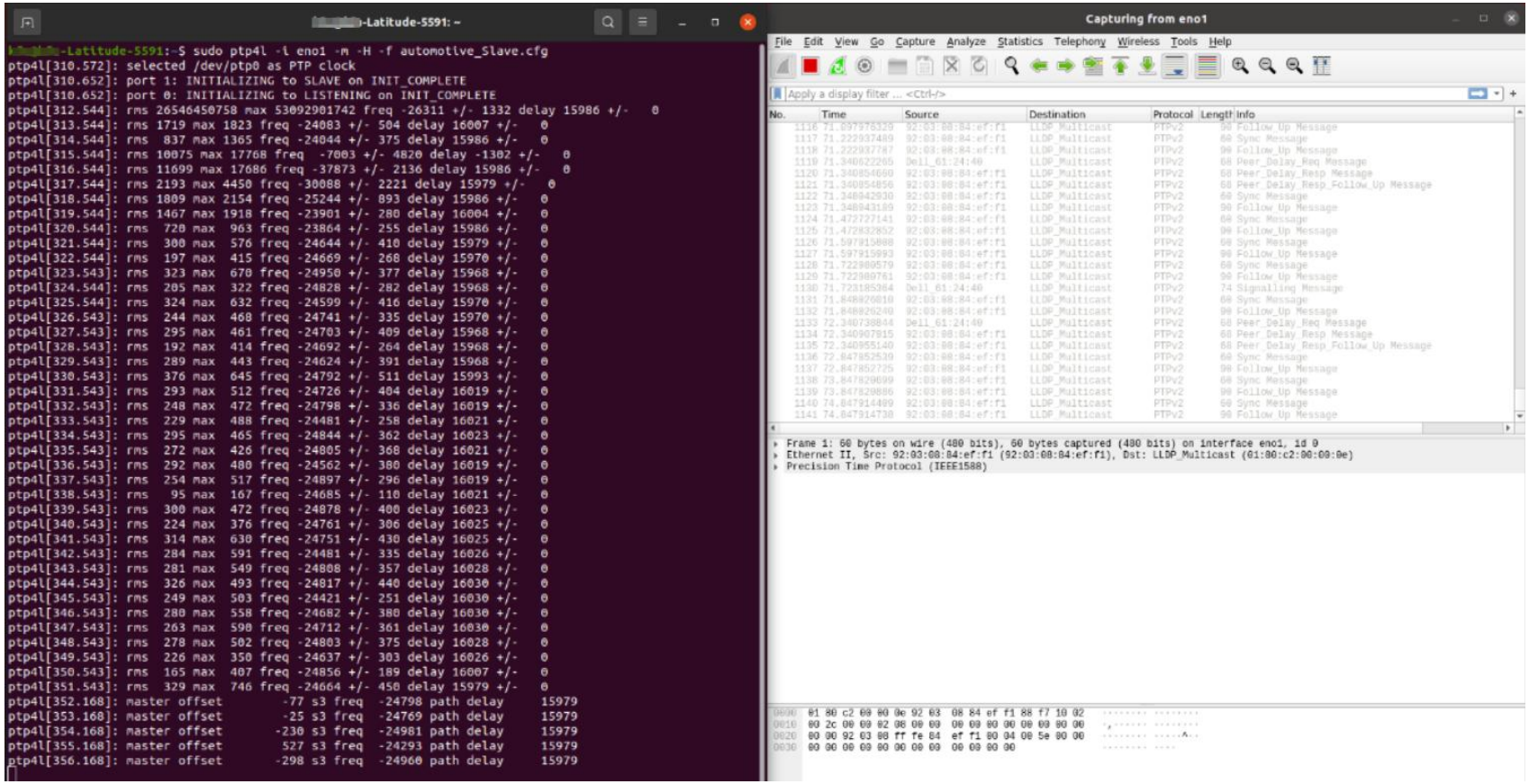
Save config to TSN Switch Flash memory

Delete config from TSN Switch Flash memory

***Only one slave clock can be sit to receive timing from an external grandmaster.**

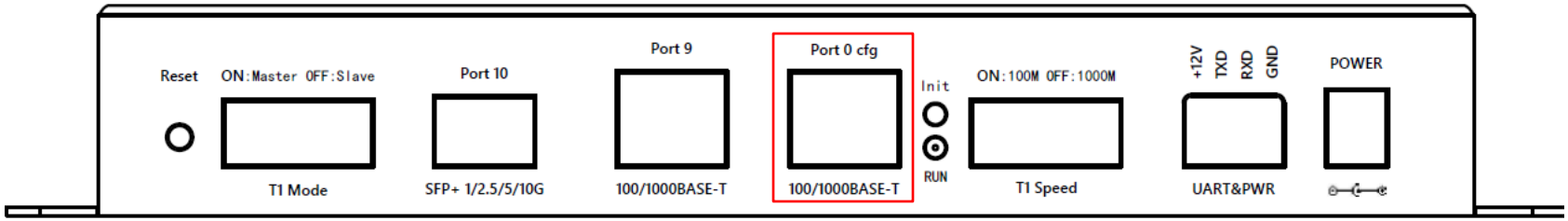
How to use SW2000TSN Config Tool

You can use a Linux PC as a slave clock endpoint. Start a linuxptp app to sync with the TSN Switch.



How to use SW2000TSN Engineering Config Tool

automotive_switch_config_4.12_win.exe



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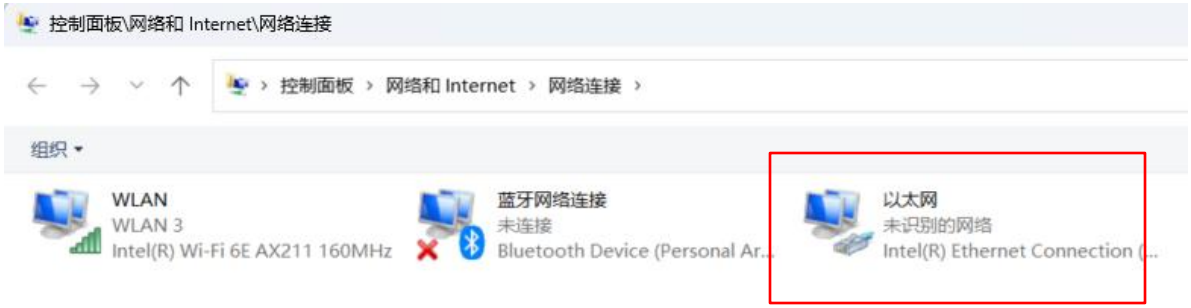

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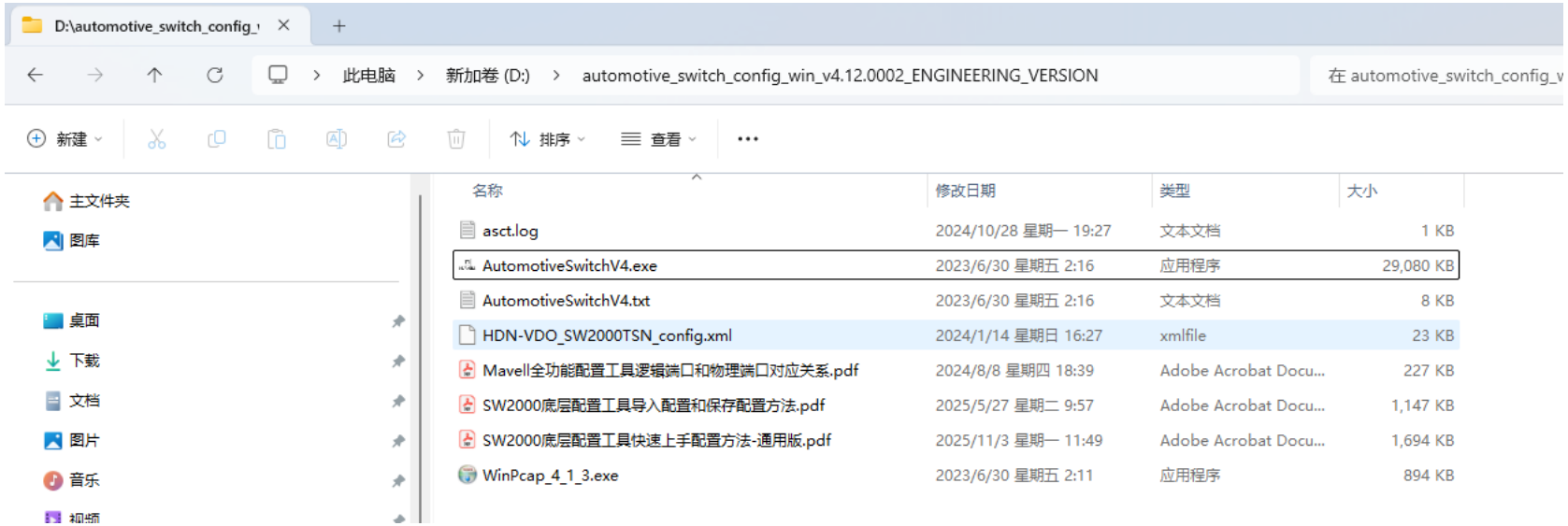
How to use SW2000TSN Engineering Config Tool

automotive_switch_config_4.12_win.exe

Please locate the path to D:\ or C:\ as below



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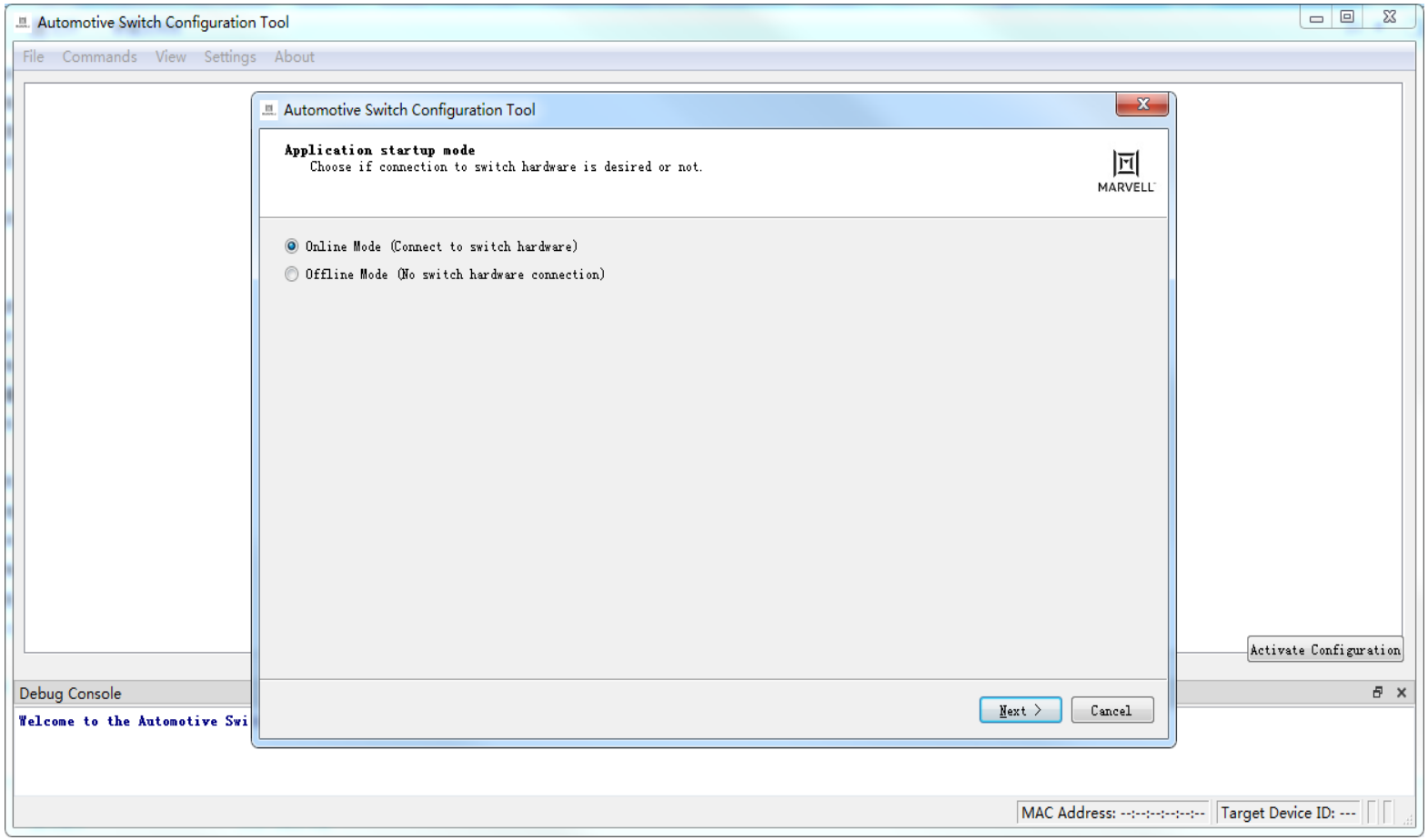
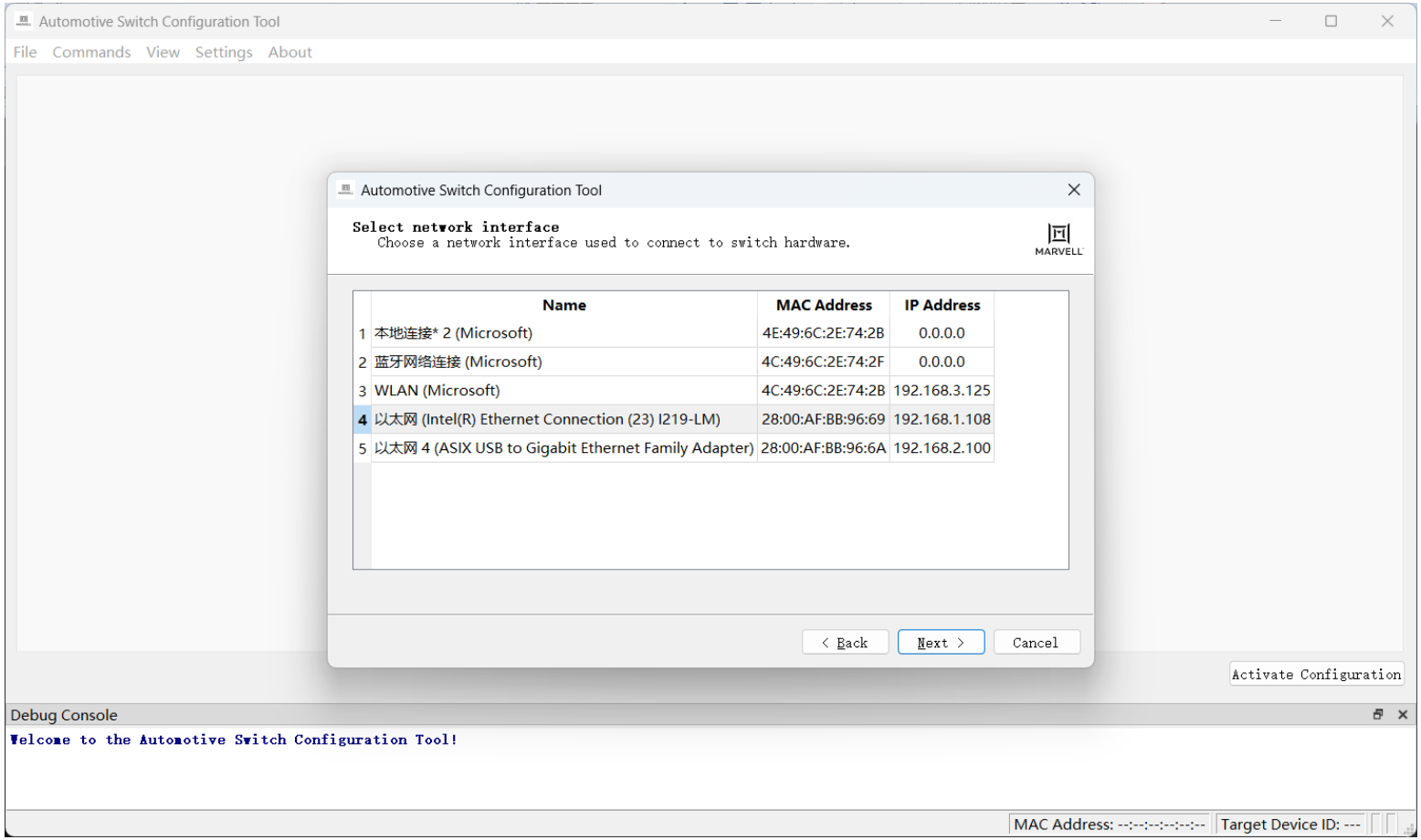
How to use SW2000TSN Engineering Config Tool

Power on TSN Switch. Double click the “AutomotiveSwitchV4.exe”, click “Online Mode”, click “Next”.



How to use SW2000TSN Engineering Config Tool

Now you will see a pop up, select the Network adapter which connect the Port0，Click “Next”



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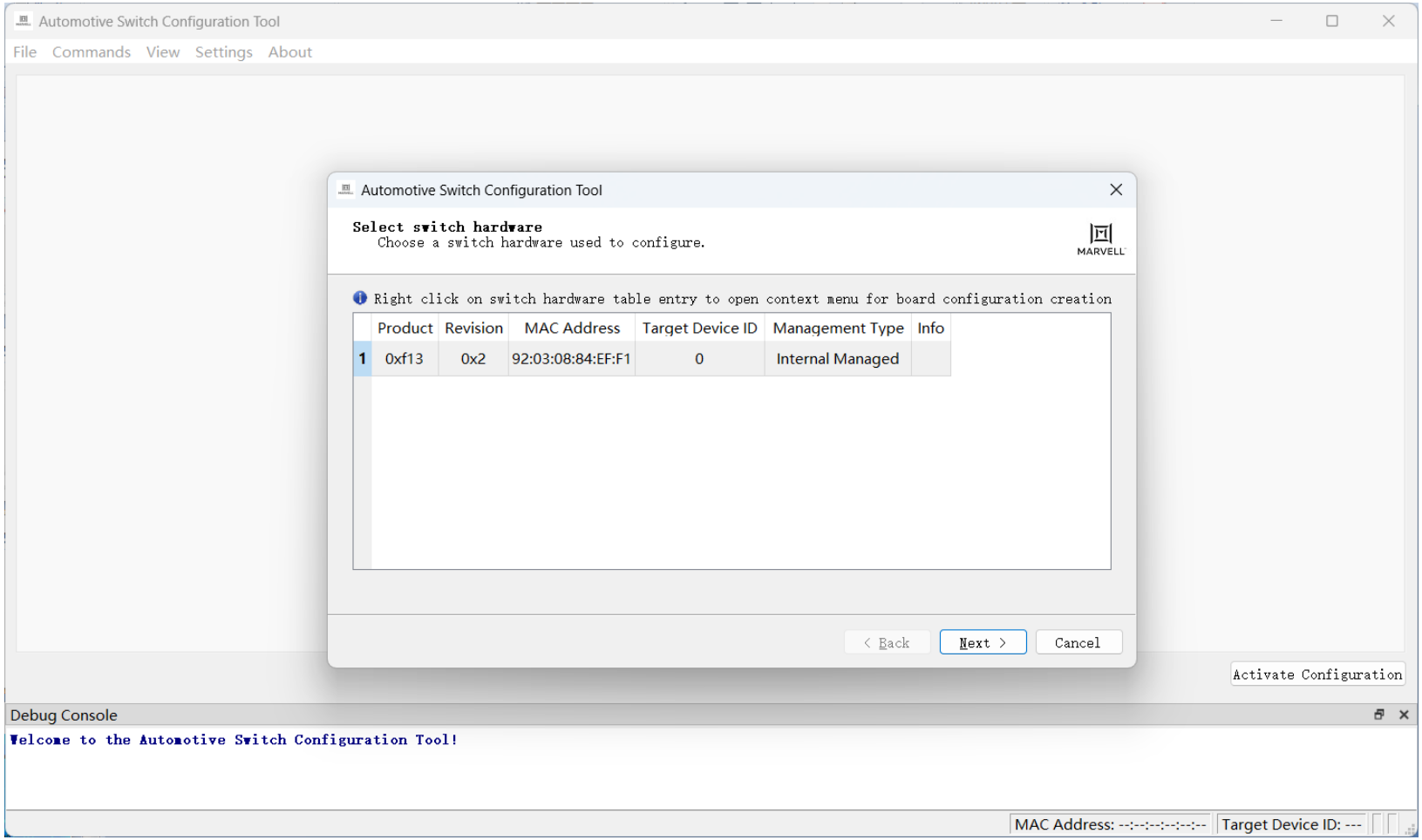
How to use SW2000TSN Engineering Config Tool

You will see a device in the list，please choose it and click “Next”.



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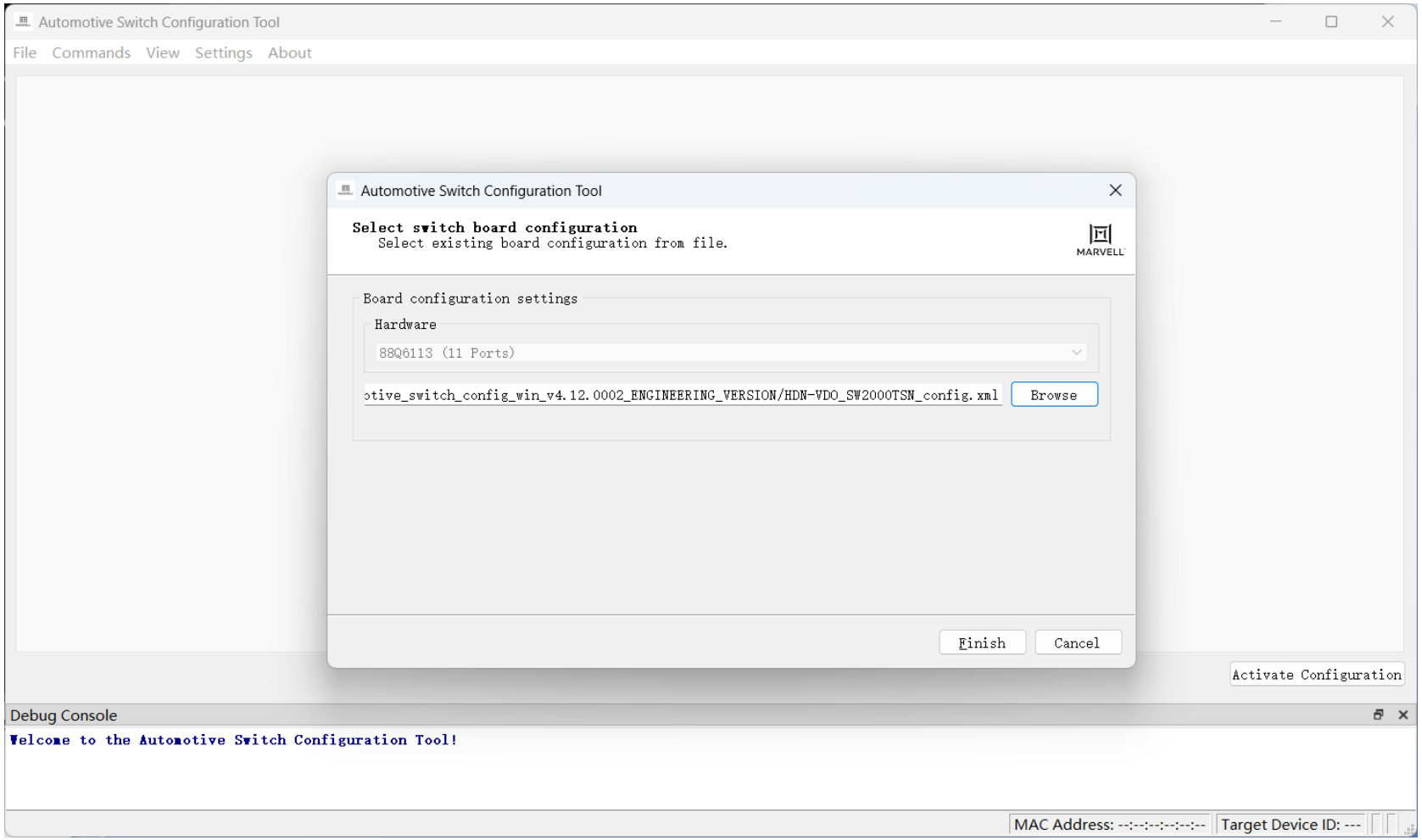
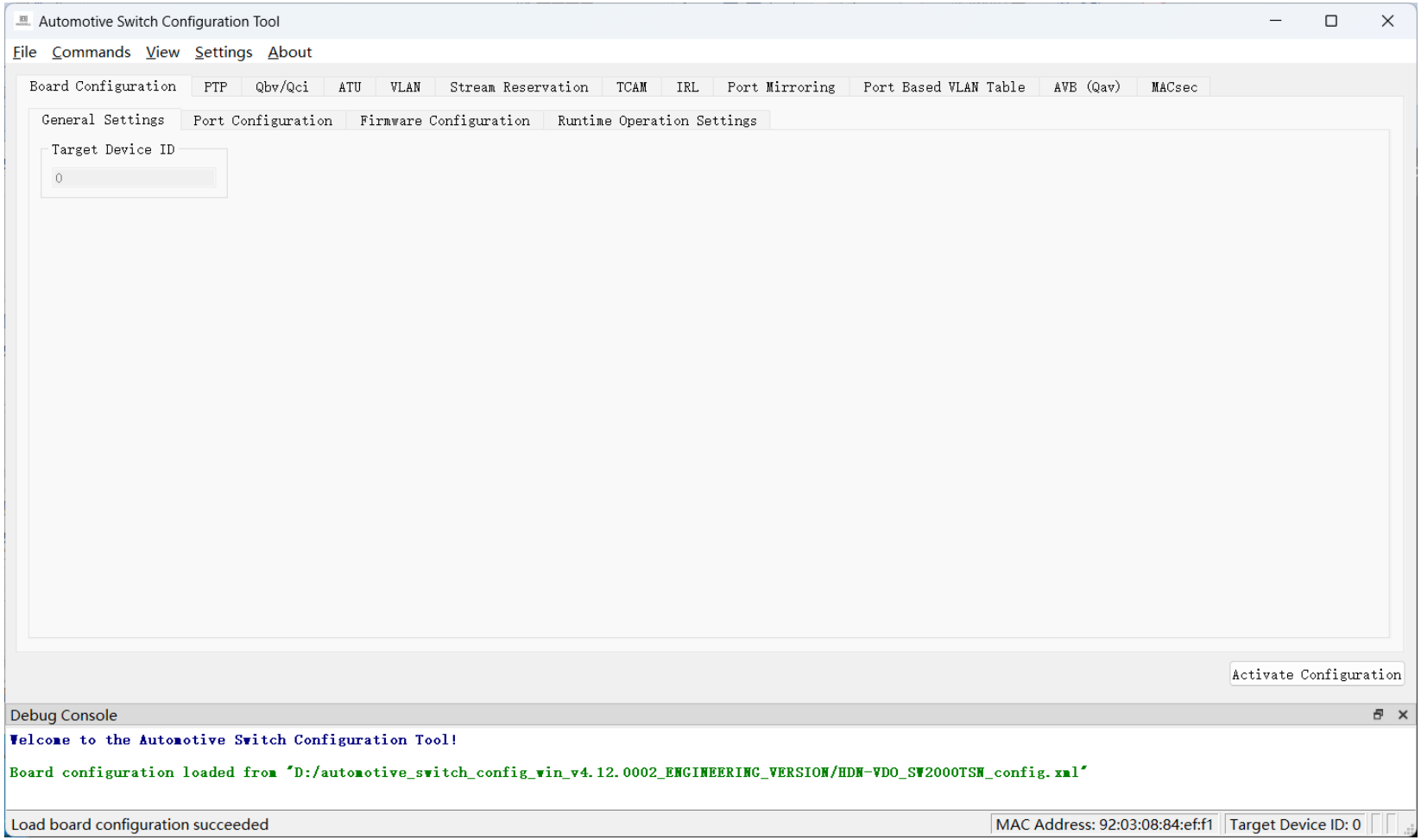
How to use SW2000TSN Engineering Config Tool

Please choose device board level config which in the software folder, named is “HDN-VDO_SW2000TSN_config.xml” ， click “Next”.



How to use SW2000TSN Engineering Config Tool

Now you will see the system config panel.



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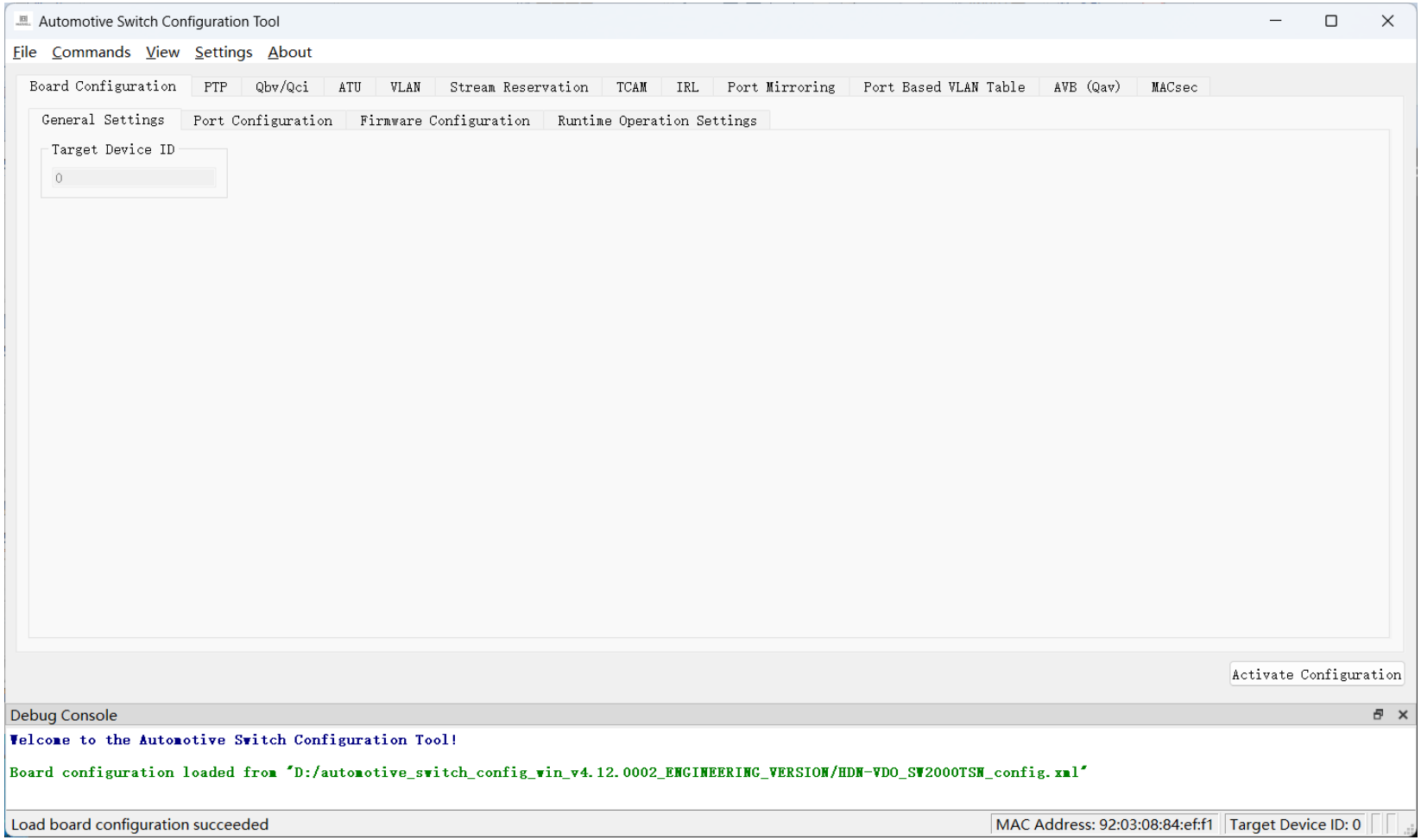

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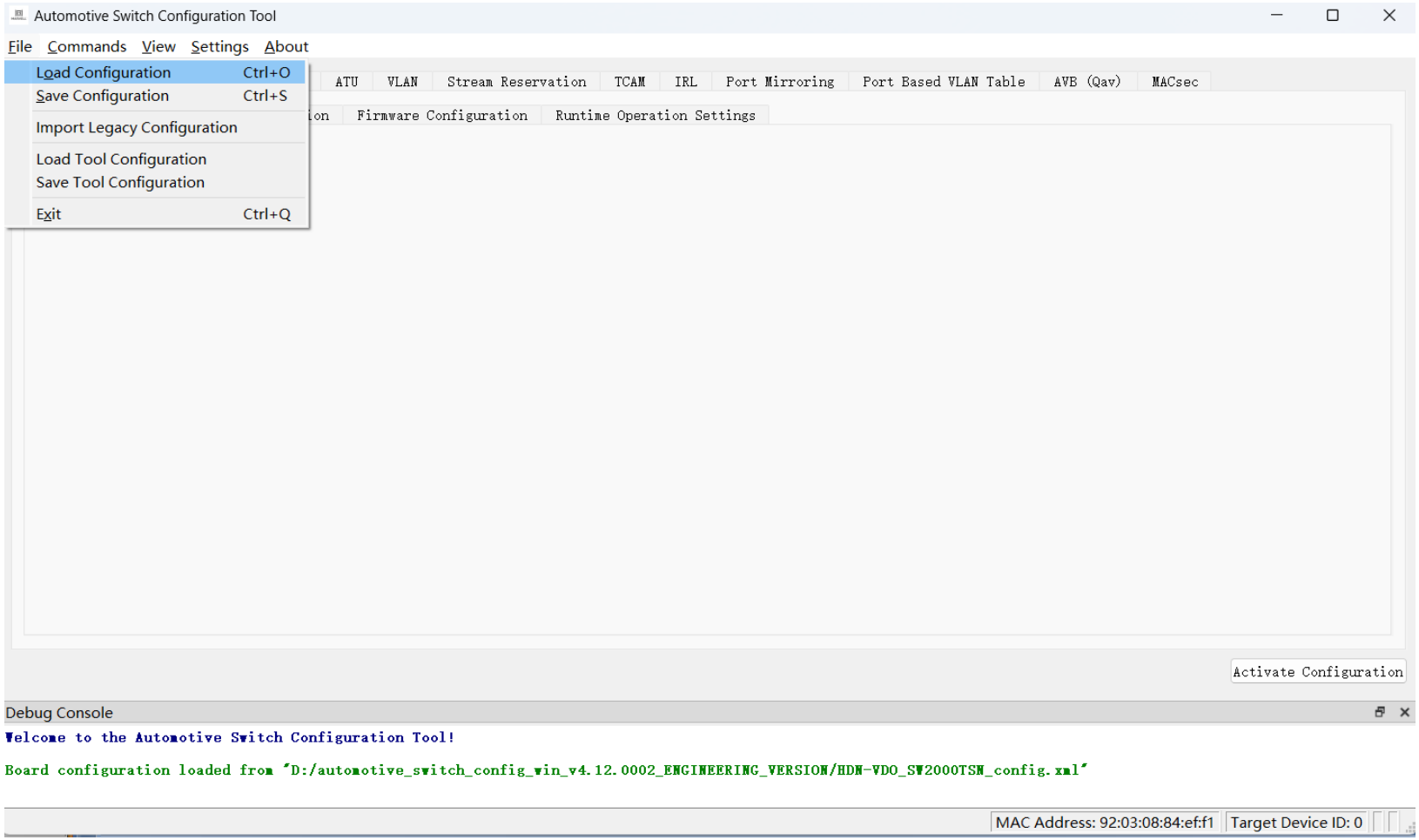
How to use SW2000TSN Engineering Config Tool

You can load the DEMO application level config file. Click top left corner "File / Load Configration", load the user config xml file.



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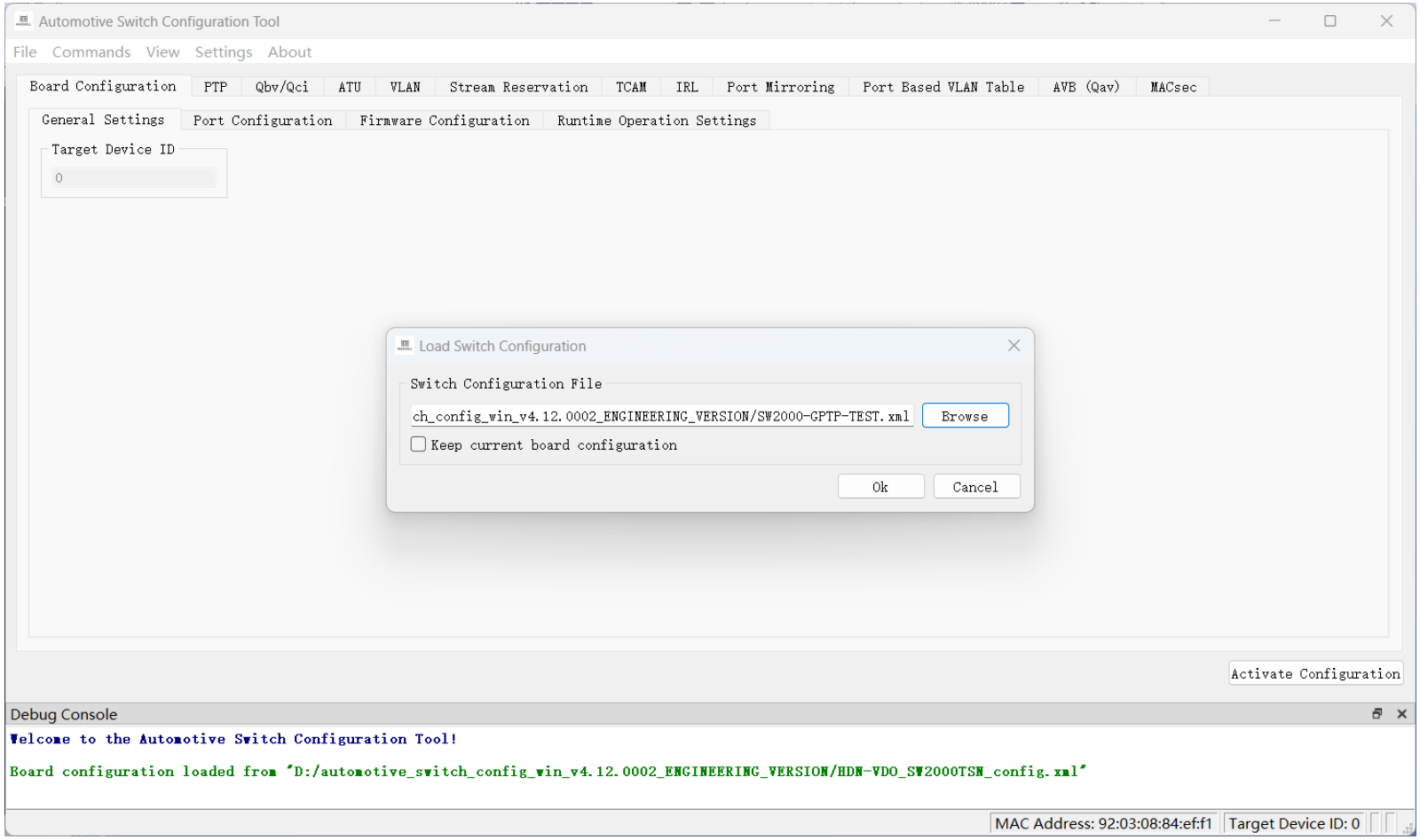
How to use SW2000TSN Engineering Config Tool

Now, go to the software path, select the Demo config xml file.



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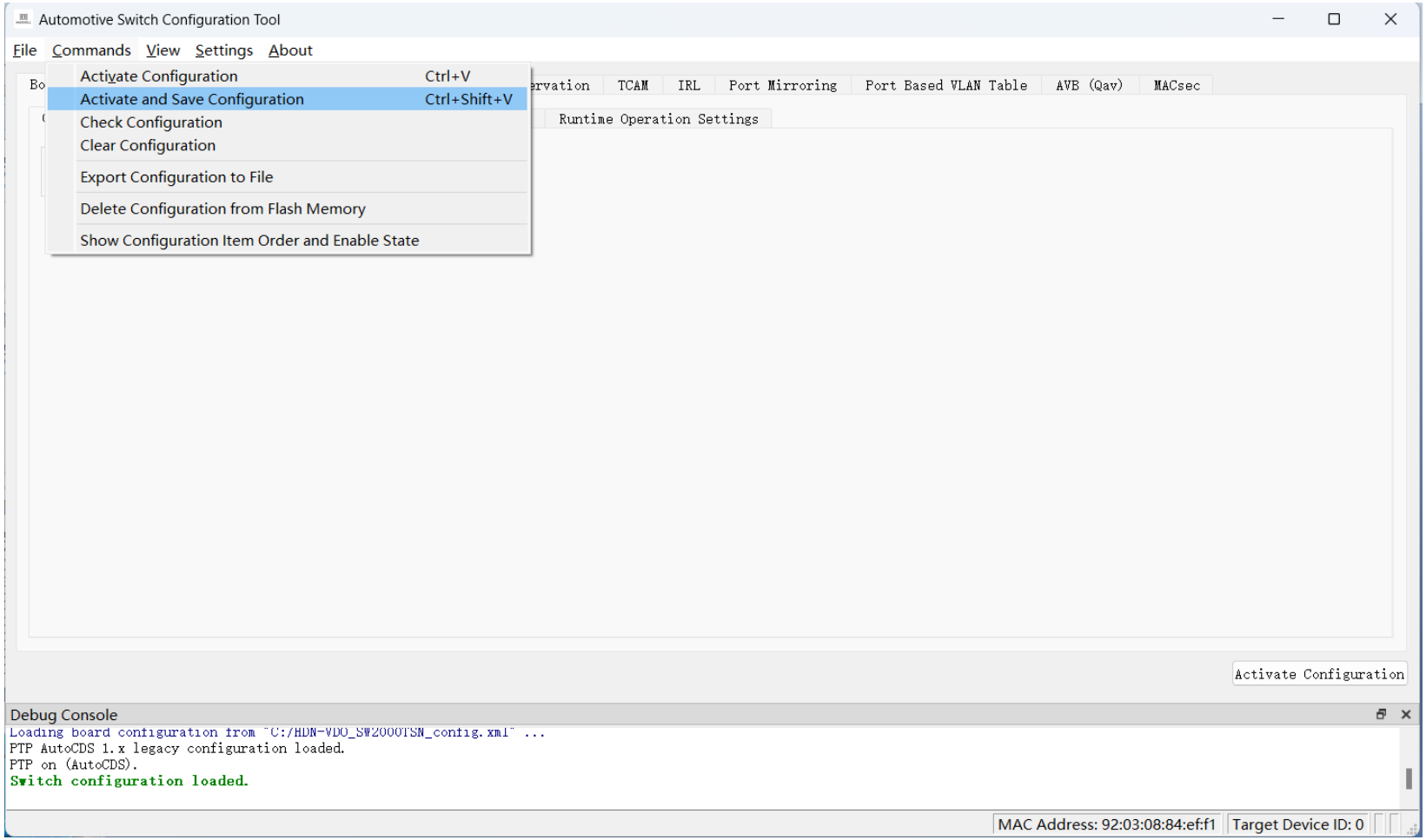
How to use SW2000TSN Engineering Config Tool

Click top left corner second item: “Conmand / Active and Save configuration”.

That will implement the config active and save.



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How to create config by Engineering Config Tool

The SW2000TSN switch port is different from the Logic port in this config tool. If you will use this tool, please refer to the map channel as below.

Logic port in config tool	Map to	SW2000TSN port	Comment
Port 0	→	For CPU, internal use	No port
Port 1	→	Port 2	T1 Port2
Port 2	→	Port 3	T1 Port3
Port 3	→	Port 4	T1 Port4
Port 4	→	Port 5	T1 Port5
Port 5	→	Port 6	T1 Port6
Port 6	→	Port 7	T1 Port7
Port 7	→	Port0	RJ45，Port0，config
Port 8	→	Port9	RJ45，Port9
Port 9	→	Port8	T1 Port8
Port 10	→	Port10	10G SFP+，Port10
Port 11	→	Port1	T1 Port1

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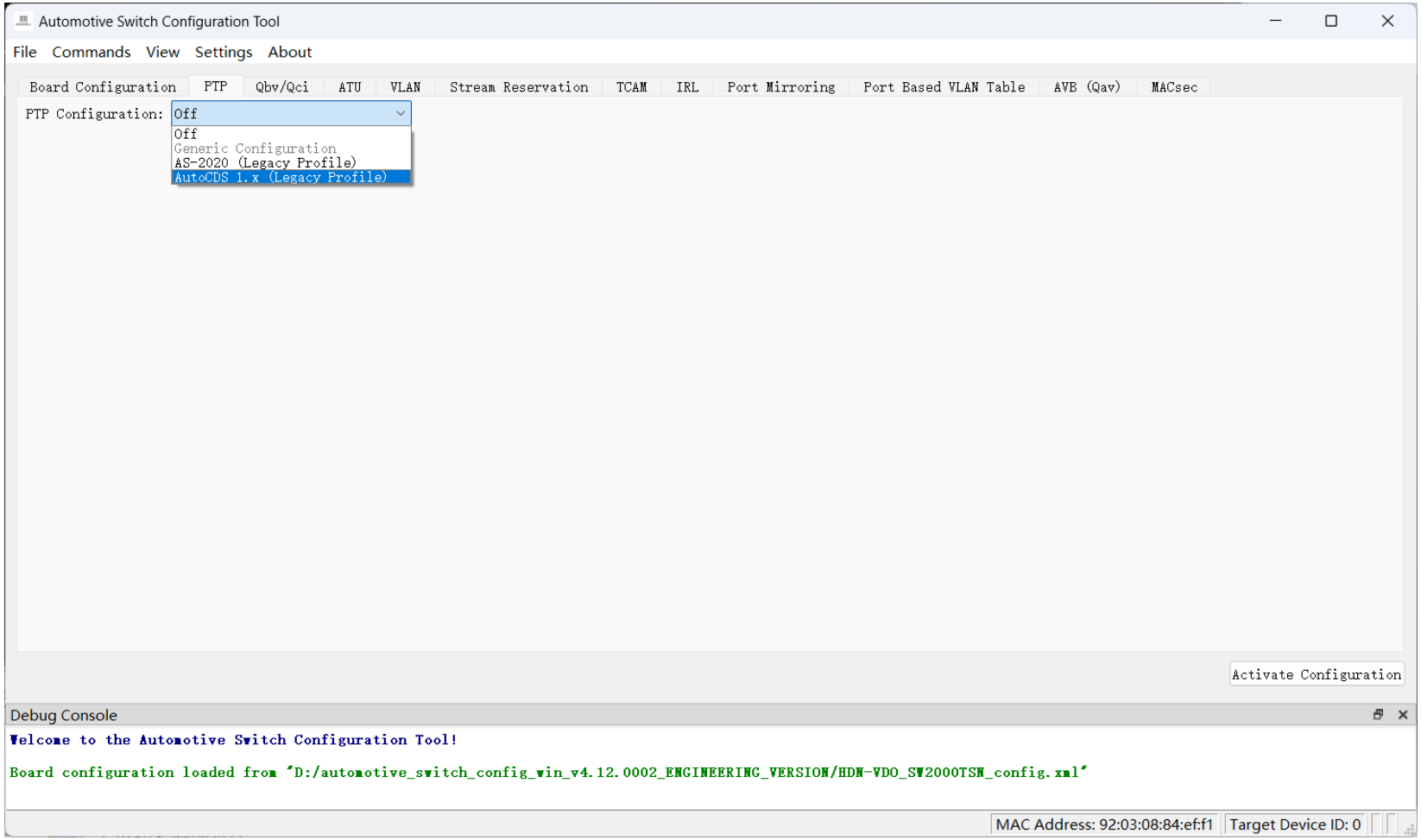
How to create config by Engineering Config Tool

This is a Domo to create a gPTP config. Port 3 is slave clock, Port 4/5/6 are master lock. Click PTP Tab.



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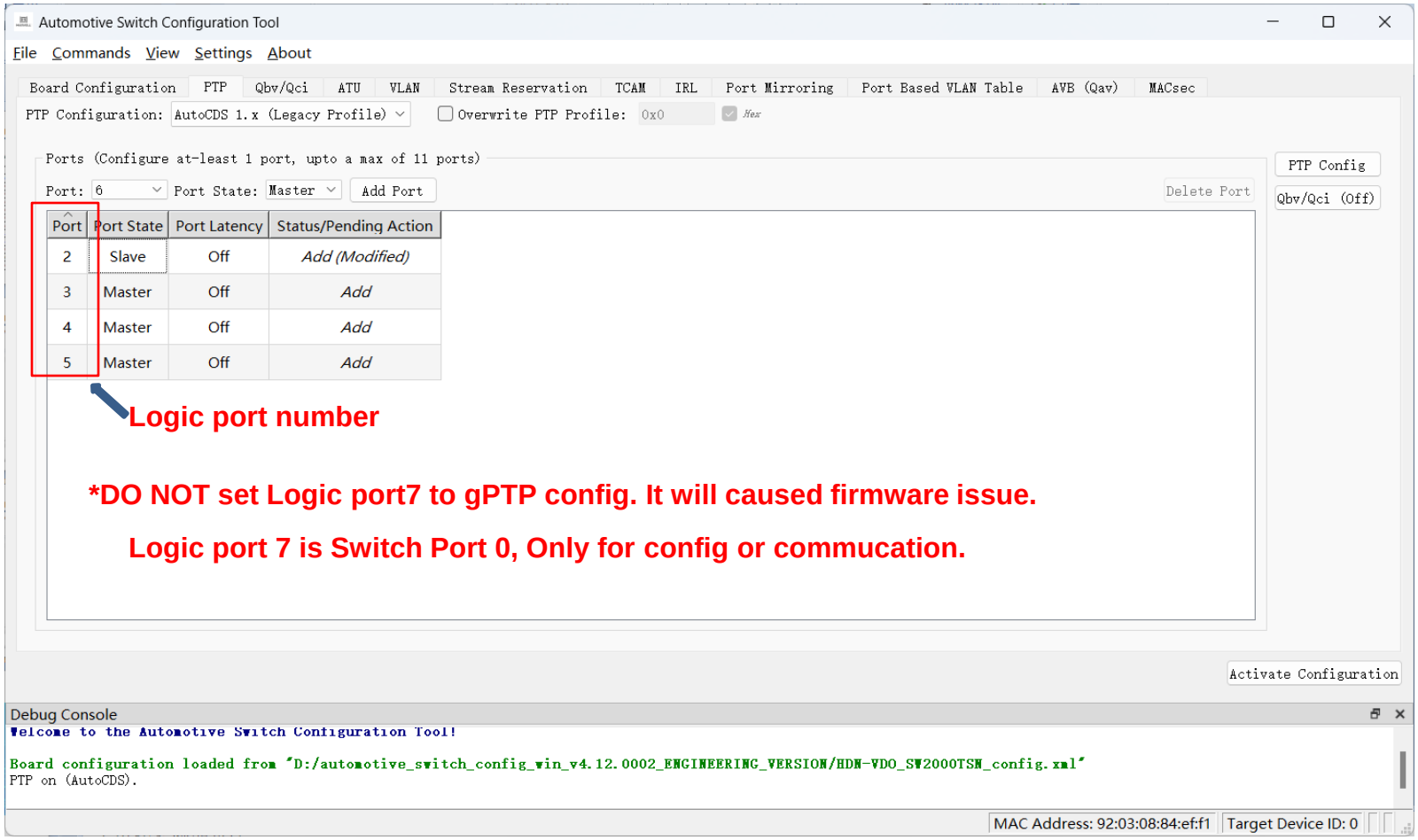
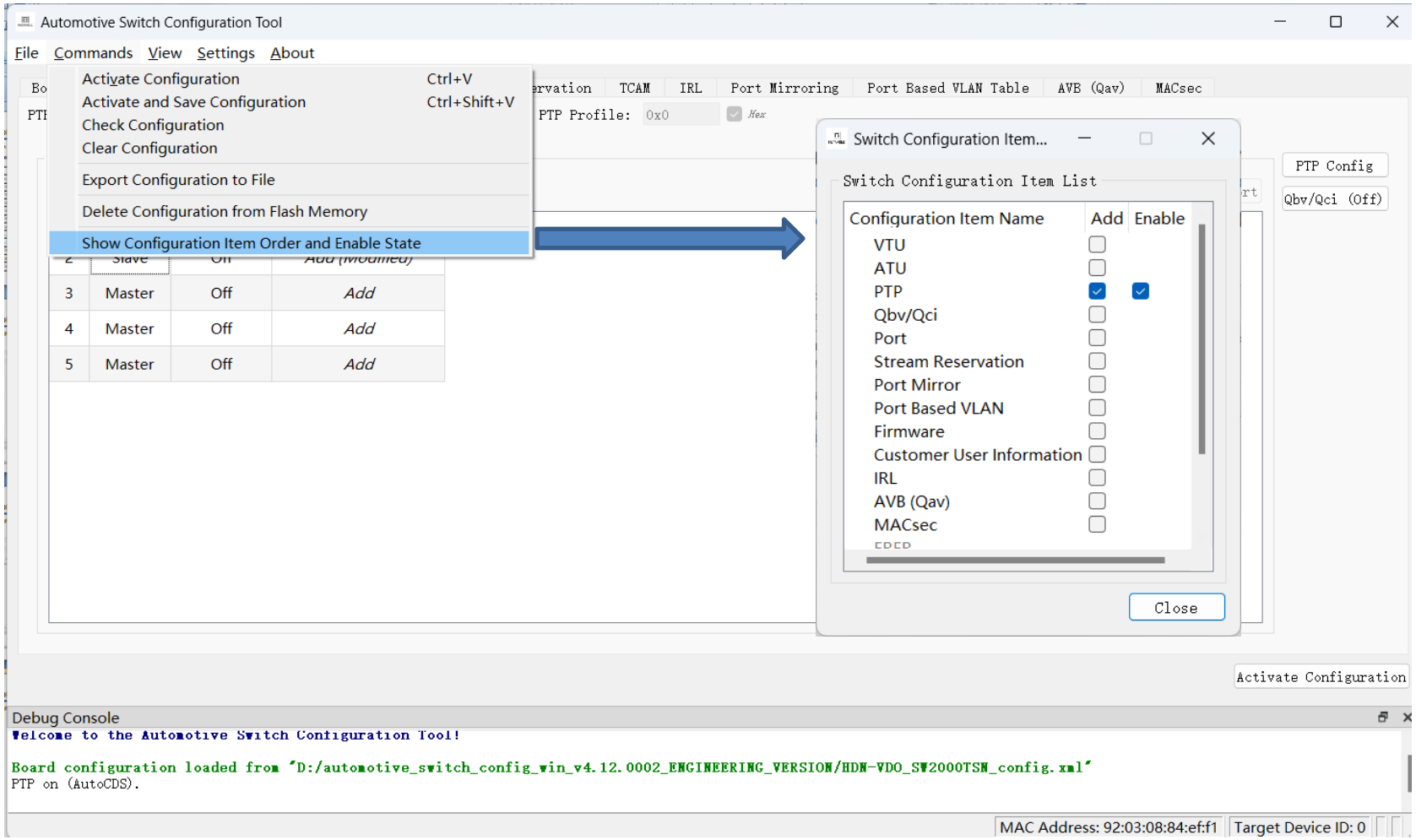


***For example, Switch Port3 is logic port2, Switch Port4/5/6 are logic port3/4/5.**

How to create config by Engineering Config Tool

Click Commands/Show Configration Item Order and Enable State.

You will see a pop up, and select PTP Add/Enable “✔”



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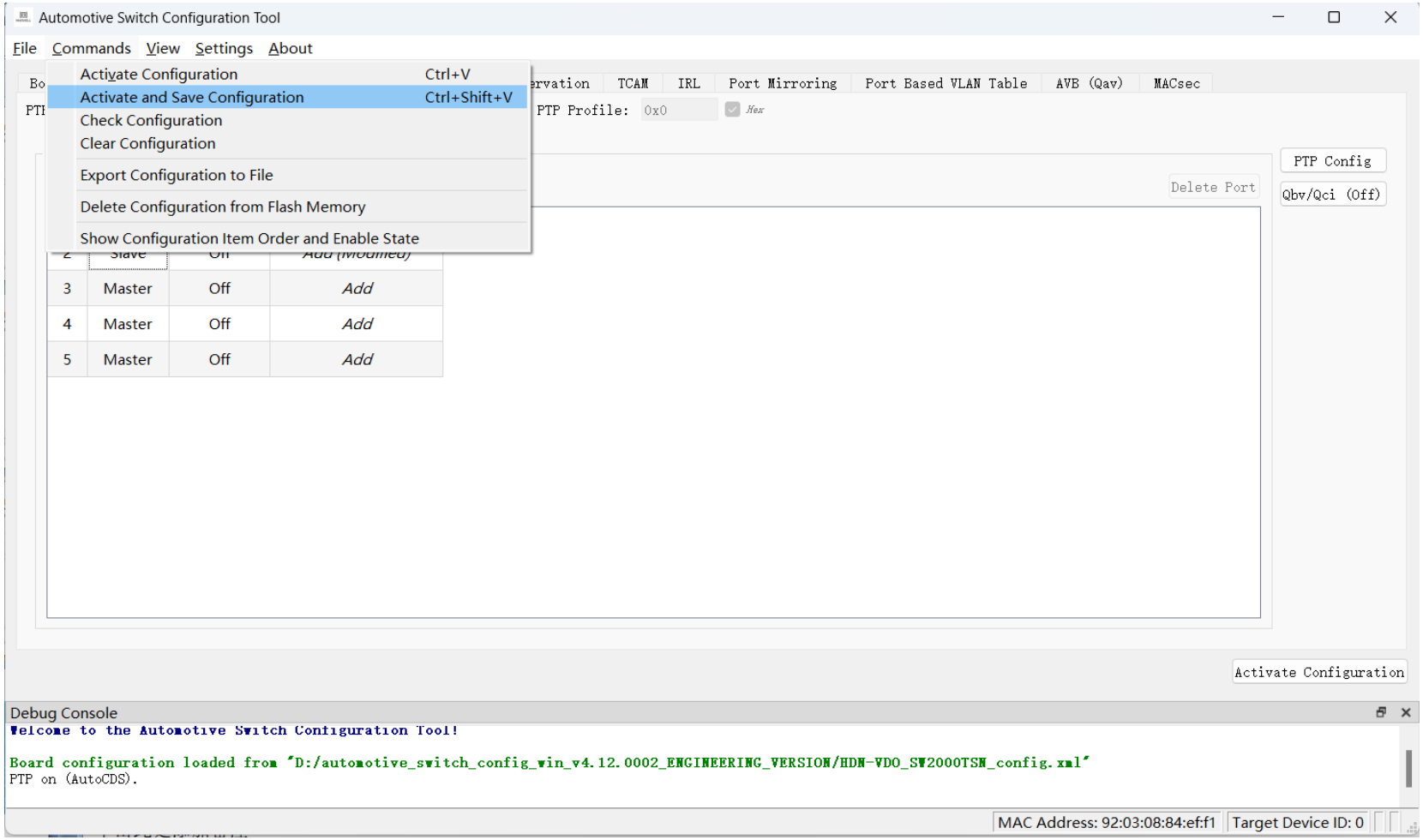
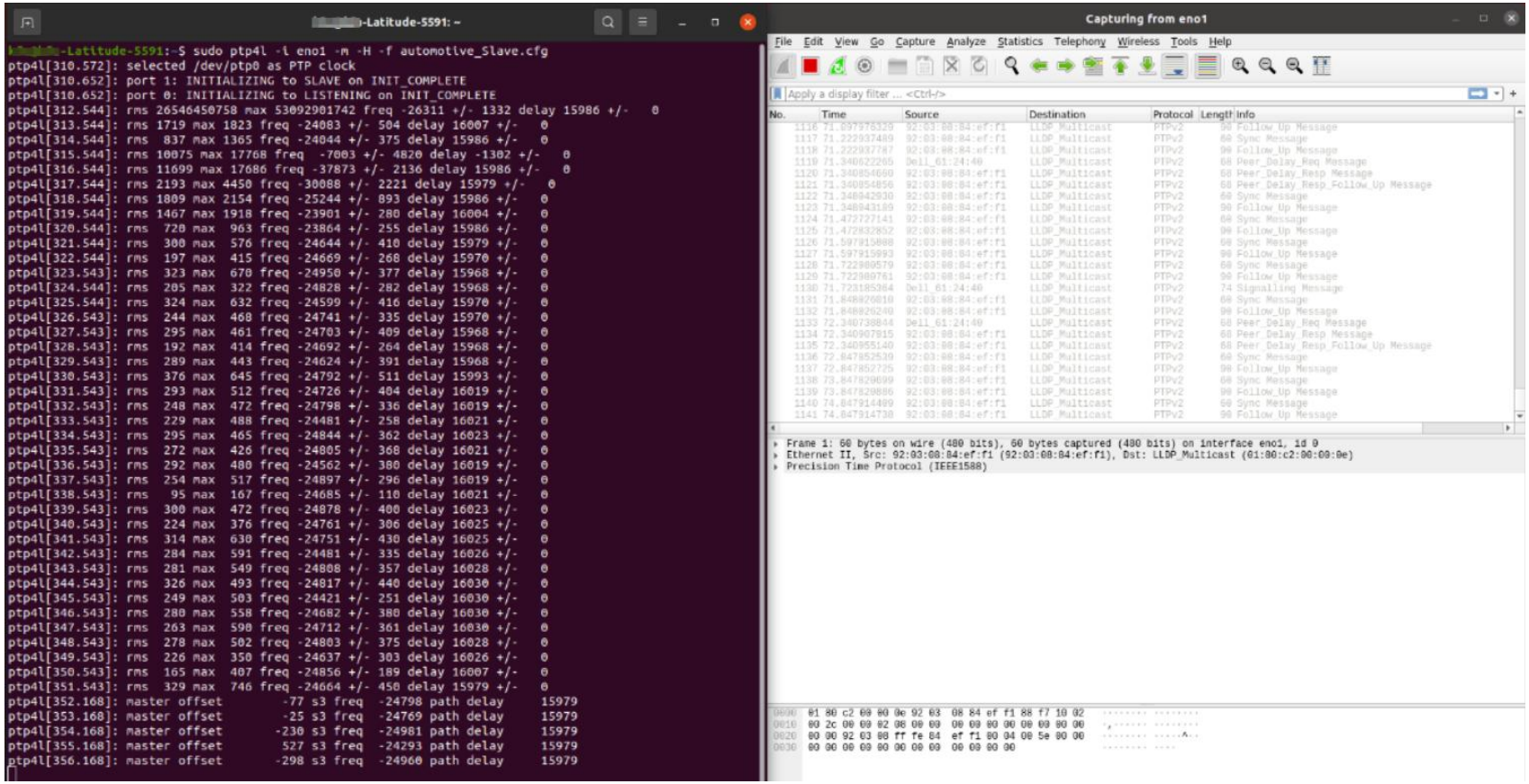
How to create config by Engineering Config Tool

You can “Active and Save Configration” to save in Flash.



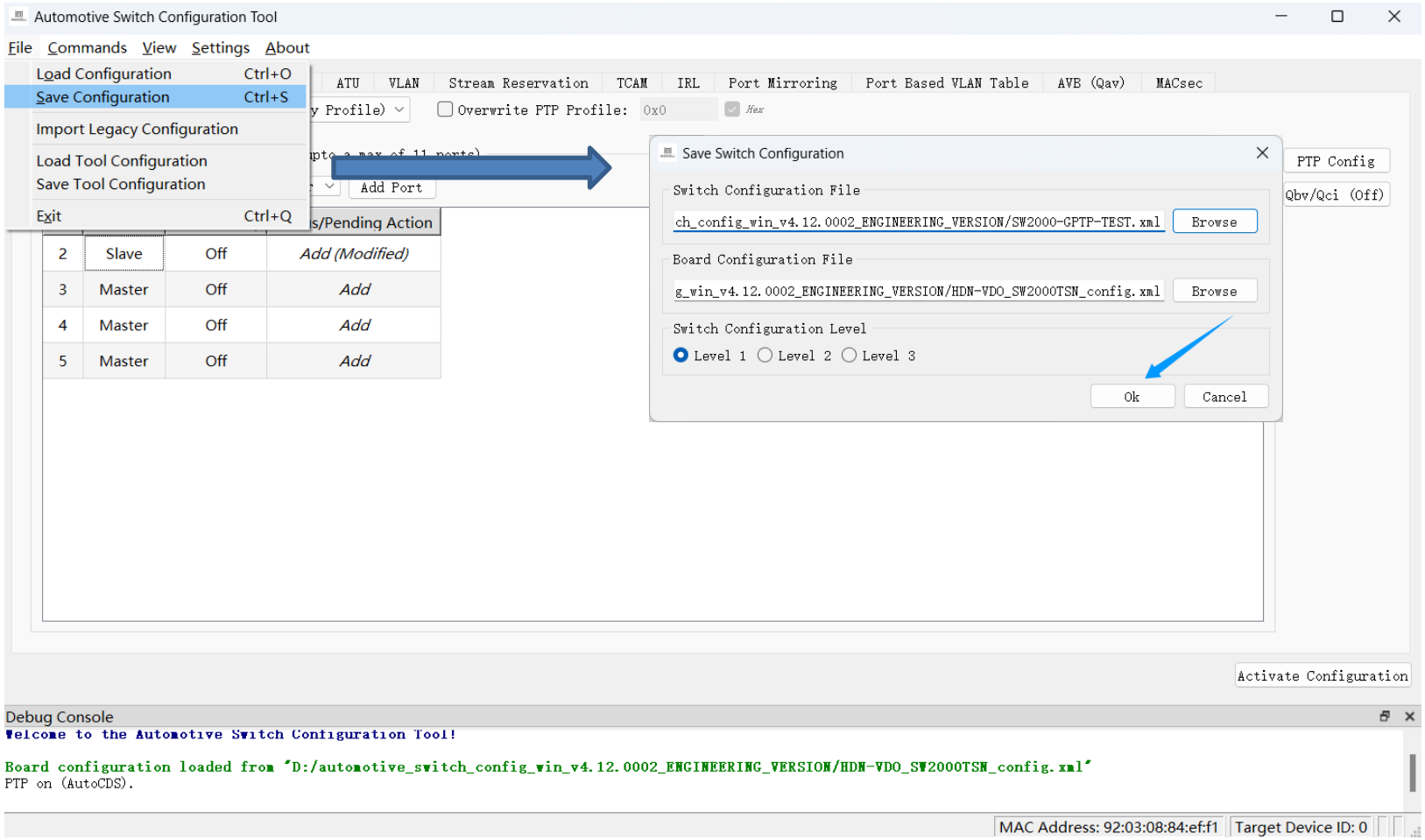
How to create config by Engineering Config Tool

You can use a Linux PC as a slave clock endpoint. Start a linuxptp app to sync with the TSN Switch.



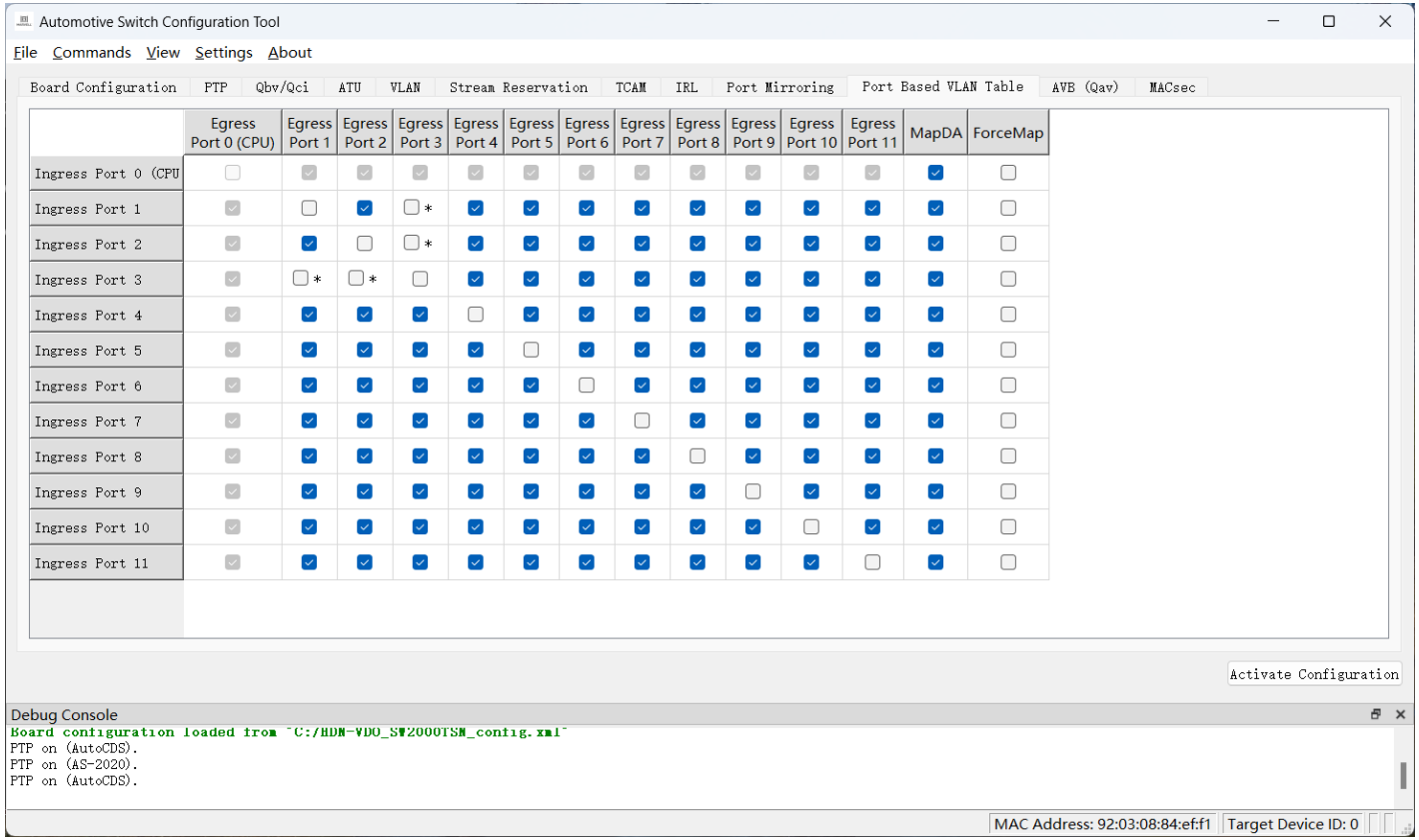
How to create config by Engineering Config Tool

You can also Save this config as a product config. Next time you just need to load this config and Save it to Flash.



How to create config by Engineering Config Tool

Port Based VLAN Table. The isolation function of the port can be achieved by unchecking the checkbox. Below, Switch Port4(logic port3) isolate with Switch Port2(logic port1) and Port3(logic port2) .



	Egress Port 0 (CPU)	Egress Port 1	Egress Port 2	Egress Port 3	Egress Port 4	Egress Port 5	Egress Port 6	Egress Port 7	Egress Port 8	Egress Port 9	Egress Port 10	Egress Port 11	MapDA	ForceMap
Ingress Port 0 (CPU	☐	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐
Ingress Port 1	☑	☐	☑	☐ *	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐
Ingress Port 2	☑	☑	☐	☐ *	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐
Ingress Port 3	☑	☐ *	☐ *	☐	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐
Ingress Port 4	☑	☑	☑	☑	☐	☑	☑	☑	☑	☑	☑	☑	☑	☐
Ingress Port 5	☑	☑	☑	☑	☑	☐	☑	☑	☑	☑	☑	☑	☑	☐
Ingress Port 6	☑	☑	☑	☑	☑	☑	☐	☑	☑	☑	☑	☑	☑	☐
Ingress Port 7	☑	☑	☑	☑	☑	☑	☑	☐	☑	☑	☑	☑	☑	☐
Ingress Port 8	☑	☑	☑	☑	☑	☑	☑	☑	☐	☑	☑	☑	☑	☐
Ingress Port 9	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐	☑	☑	☑	☐
Ingress Port 10	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐	☑	☑	☐
Ingress Port 11	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☐	☑	☐

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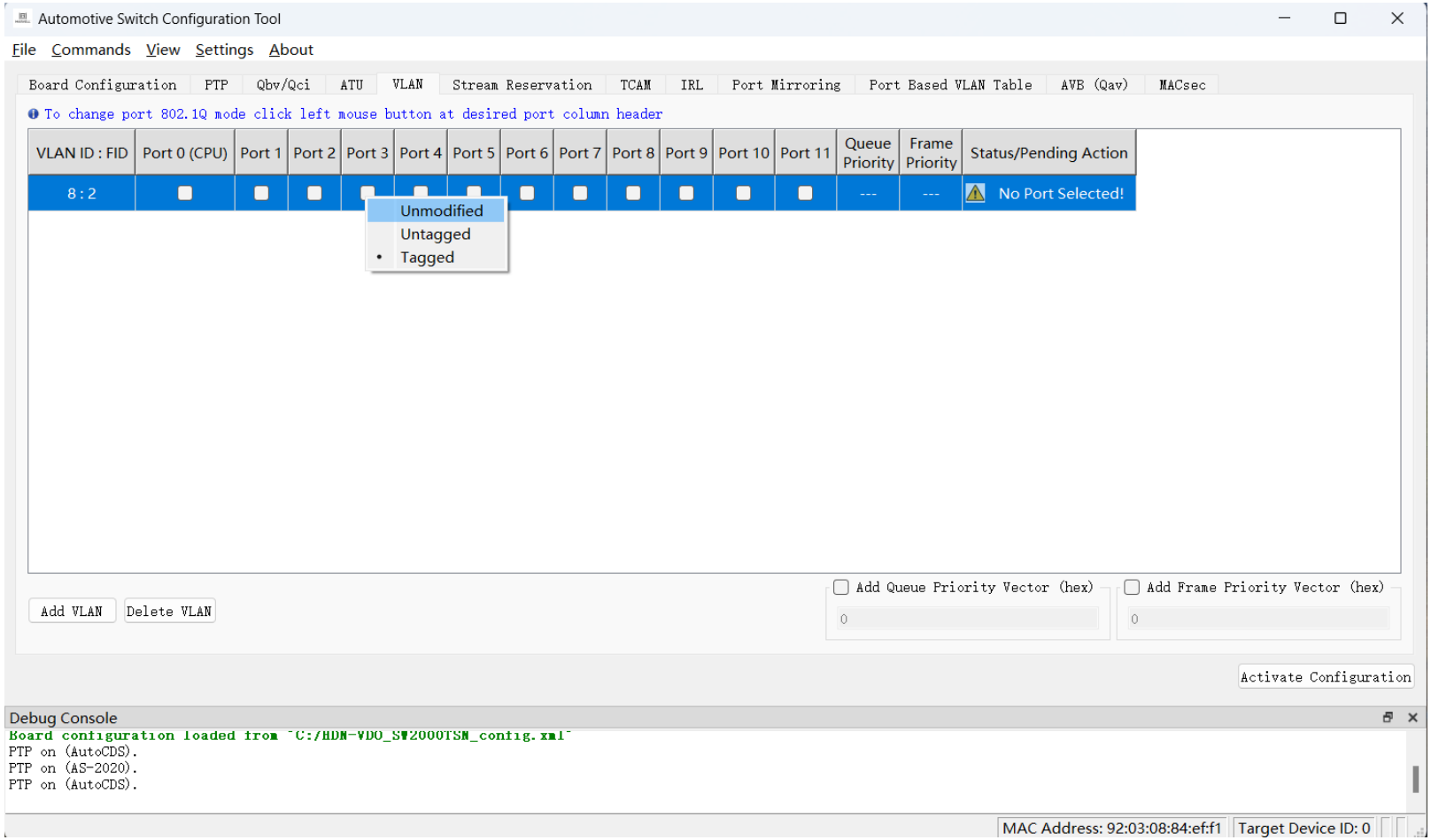
How to create config by Engineering Config Tool

VLAN Table. The switch supports internal VLAN table.



If you want to set VLAN table and PVID, please reach out to us, we can send you a DEMO. It enables you to implement applications faster.

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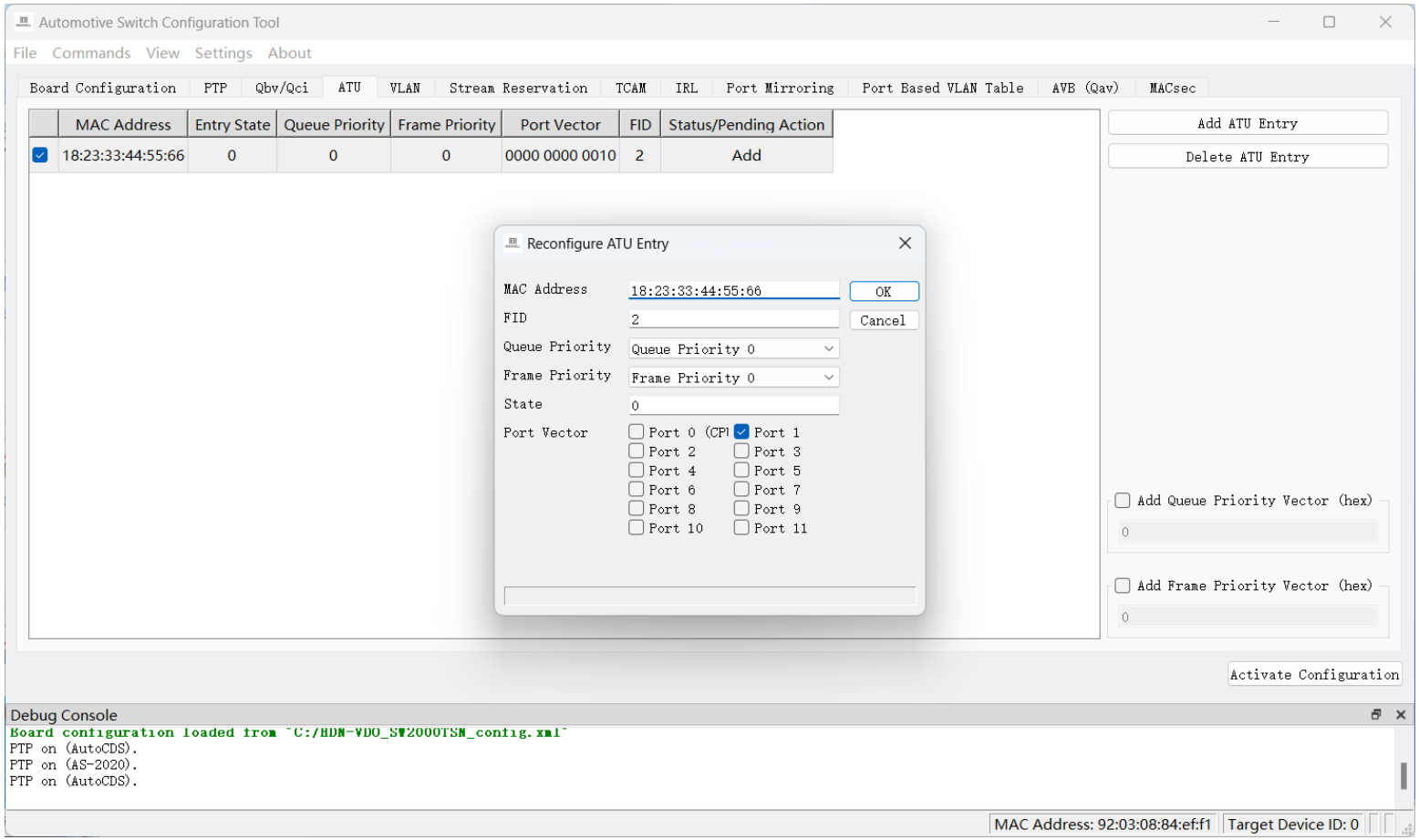
How to create config by Engineering Config Tool

ATU Table. The switch supports MAC binding via ATU.



Above setting, Switch Port2(logic port 1) supports the input of data from devices with a MAC address of 18:23:33:44:55:66, others will drop.

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THANK YOU!

谢谢您！

联系我们：17265076798

重庆汇迪能智能技术有限公司